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UEI Architecture and the Digital Energy Grid

Enabling the Transition from Fragmented Platforms to a Unified Energy Ecosystem

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Solving Issues of Fragmented Digital Ecosystems

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Today's Reality

Energy services operate as **digital islands**

Proprietary platforms and vendor lock-in

Consumers must navigate **multiple disconnected apps**

Connected Ecosystem (UEI & IES)

The Core Barrier

No common **service discovery language** for energy systems

UEI — The Language of the Digital Energy Grid

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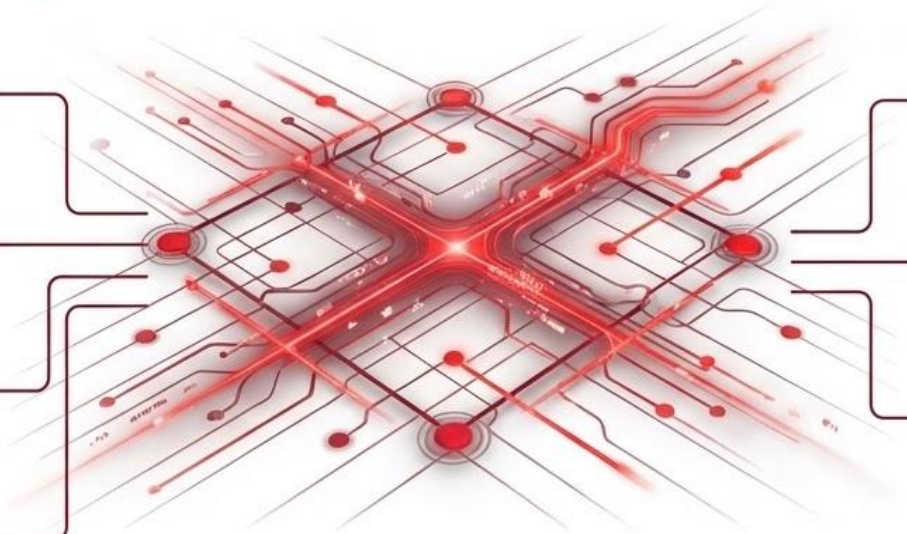
UEI defines a common protocol layer for energy services.

It enables interoperability between:

- Consumers
- Utilities
- Energy service providers
- Distributed Energy Resources (DER)

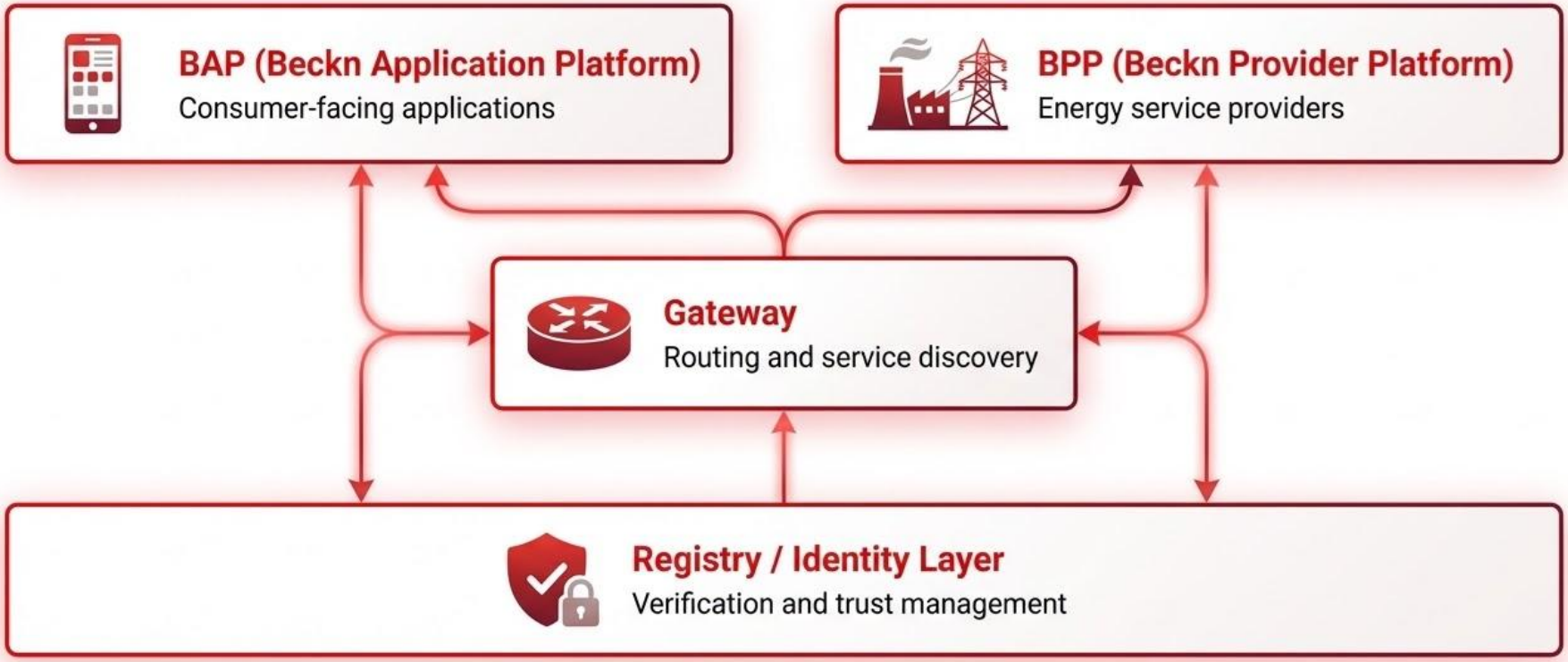


Core Functions

- Service discovery
- Transaction orchestration
- Cross-platform settlement

The Architecture of Trust (Beckn)

UEI uses the open-source Beckn Protocol, enabling decentralized digital networks.



UEI Digital Energy Architecture

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UEI Architecture: Bridging Physical and Digital Energy Systems

UEI connects energy infrastructure with digital services.

Energy Applications	Demand response. Energy trading, EV charging services, Smart city platforms
UEI Interface Layer	Discovery, Transaction orchestration, Interoperability,
Communication Layer	Wi-SUN RF Mesh. NB-IoT, Cellular LTE, PLC. LoRaWAN
Grid Devices	Smart meters, Solar inverters, V chargers, Battery storage

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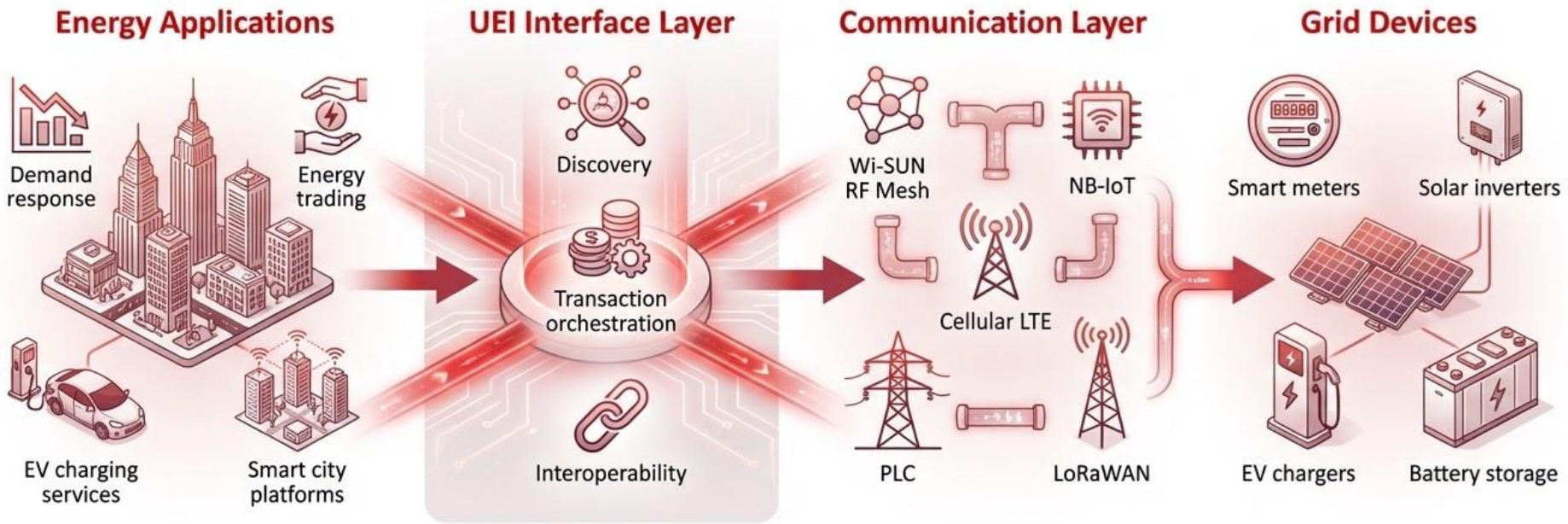

Unified Energy Service Discovery : One Click Total Access

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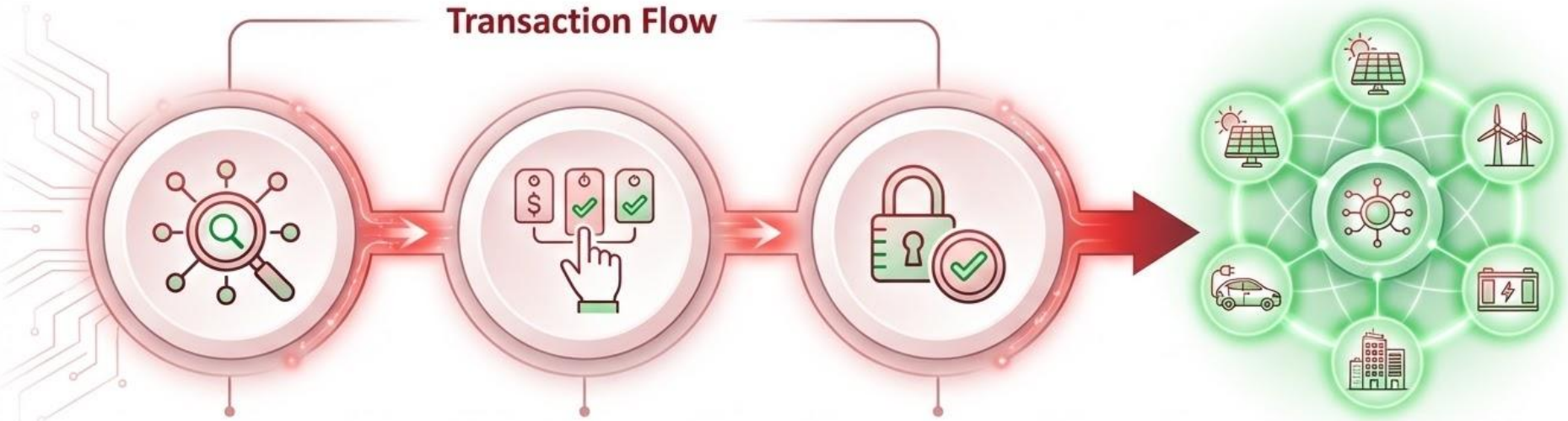
UEI enables seamless discovery of energy services.



Search

Discover energy services across multiple providers.

Select

Compare real-time availability and pricing.

Settle

Execute secure digital payment and confirmation.

Outcome

A decentralized energy marketplace.

The Digital Nervous System of the Grid

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From Transactions to Grid Intelligence

When millions of grid devices connect through UEI. the energy system becomes:

- **Observable**
- **Programmable**
- **Data-driven**

UEI enables:

- Automated demand response
- Distributed grid orchestration
- Real-time grid analytics
- Smart meters and IoT sensors provide **continuous telemetry**.

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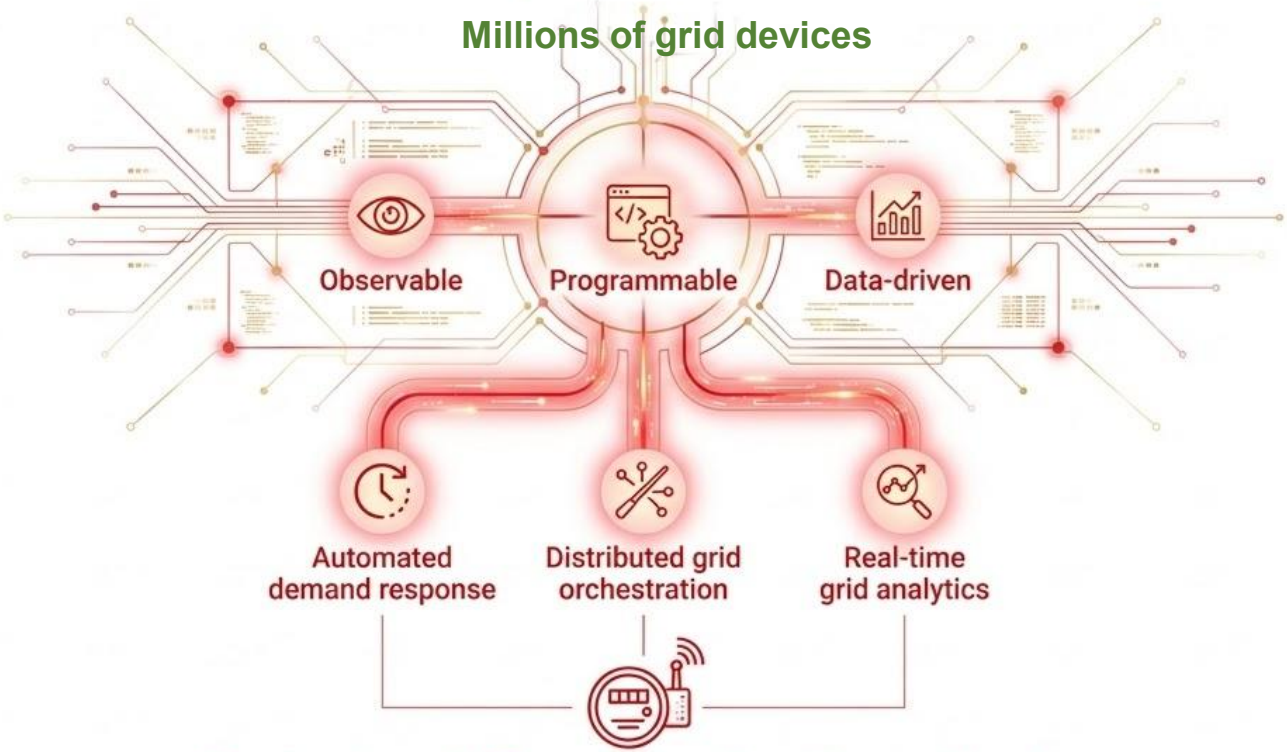

Roadmap: The Internet of Energy

From Pilot Networks to National Energy Infrastructure

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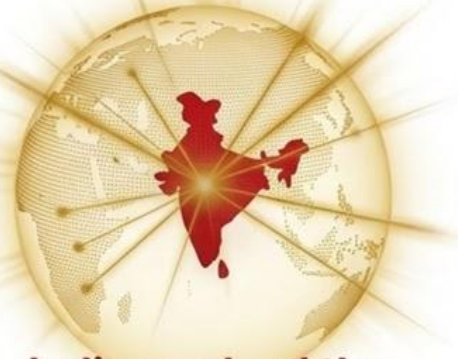
Next steps for UEI adoption:

Deploy UEI pilot networks

Integrate utilities and DER operators

Expand across EV charging and renewable ecosystems

Strategic Vision



India can lead the world in building the Internet of Energy.

Translating UEI Architecture into Grid Operations

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Eppeltone HES: Upcoming UEI-Ready Grid Infrastructure

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HES enables large-scale device orchestration and real-time grid visibility.

Core capabilities:

- Integration with smart meters and grid IoT devices
- Secure communication with millions of endpoints
- Real-time telemetry and event management
- Scalable architecture for AMI and smart grid deployments

UEI Compatibility:

- Enables grid asset discovery for UEI services
- Provides real-time energy availability data
- Supports secure integration with digital energy platforms

Eppeltone HES acts as the operational bridge between physical grid infrastructure and UEI-based digital energy services.

Eppeltone Multi-Network Communication Module

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Reliable communication infrastructure is critical for UEI-enabled energy ecosystems.

Multi-Network Connectivity:

- RF Mesh (Wi-SUN)
- NB-IoT / Cellular LTE
- PLC communication
- Hybrid communication architectures

Key Capabilities:

- Secure device authentication
- OTA firmware upgrades
- High reliability for urban and rural deployments

Key Message: Eppeltone communication modules ensure that grid devices remain continuously connected to the digital energy ecosystem.

Peer-to-Peer Energy Markets on UEI Infrastructure

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UEI enables decentralized energy marketplaces where distributed resources and millions of devices participate directly.

Eppeltone P2P energy enablement capabilities:

- Meter-verified energy measurement
- Real-time energy availability signals
- Secure transaction data for settlement
- Integration with distributed energy resources & million devices

R&D Status: Beta platform under development – expected by Q2 of 2026.

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THANK YOU

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Energy Applications

- Demand response
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- EV charging services
- Smart city platforms

UEI Interface Layer

- Discovery
- Transaction orchestration
- Interoperability

Communication Layer

- Wi-SUN RF Mesh
- NB-IoT
- Cellular LTE
- PLC
- LoRaWAN

Grid Devices

- Smart meters
- Solar inverters
- EV chargers
- Battery storage